



Stochastic Processes

Lecture 18

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On Recurrence Times

- For $x, y \in \mathcal{X}$, let

$$f_{xy}^{(n)} := \mathbb{P}\left(\{\tau_y^{(1)} = n\} \mid \{X_1 = x\}\right), \quad n \in \{2, 3, \dots\}.$$

Interpretation of $f_{xy}^{(n)}$: **probability of first hit to state y at time n when starting in state x**

- For $x, y \in \mathcal{X}$, let

$$f_{xy} = \mathbb{P}\left(\{\tau_y^{(1)} < +\infty\} \mid \{X_1 = x\}\right) = \sum_n f_{xy}^{(n)}.$$

- Interpretation of f_{xy} : **probability of ever hitting state y conditioned on $\{X_1 = x\}$**
- Interpretation of $(1 - f_{xy})$: **probability of never hitting state y conditioned on $\{X_1 = x\}$**

- For any $x, y \in \mathcal{X}$,

$$f_{xy}^{(n)} \leq P^n(x, y).$$

Recurrent and Transient States

Definition (Recurrent and Transient States)

A state $x \in \mathcal{X}$ is called **recurrent** if $f_{xx} = 1$.

A state $x \in \mathcal{X}$ is called **transient** if $f_{xx} < 1$.

Remarks:

- The PMF

$$\left[f_{xx}^{(2)}, f_{xx}^{(3)}, \dots, 1 - f_{xx} \right]$$

is called **first recurrence time distribution**

- One can then ask: what is the mean under the above PMF?

Mean Recurrence Time

Definition (Mean Recurrence Time)

The **mean recurrence time** of a state $x \in \mathcal{X}$ is denoted by μ_{xx} and is defined by

$$\mu_{xx} := \mathbb{E} \left[\tau_x^{(1)} \mid \{X_1 = x\} \right].$$

Remarks:

- If x is transient, then $\mu_{xx} = +\infty$
- If x is recurrent, then

$$\mu_{xx} = \sum_{n \geq 2} n f_{xx}^{(n)}.$$

- If $\mu_{xx} < +\infty$, then x is called a **positive recurrent** state
- If $\mu_{xx} = +\infty$, then x is called a **null recurrent** state
- A state x can be only one of 3 kinds: transient, positive recurrent, or null recurrent

Visitation Probabilities

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 1

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 1

For any $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited **finitely many times**}\} \mid \{X_1 = x\}\right) = \begin{cases} 1, & f_{yy} < 1, \\ 1 - f_{xy}, & f_{yy} = 1. \end{cases}$$

Corollary 2

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 2

For any $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited infinitely many times}\} \mid \{X_1 = x\}\right) = \begin{cases} 0, & f_{yy} < 1, \\ f_{xy}, & f_{yy} = 1. \end{cases}$$

Corollary 3

Theorem (Visitation Probabilities)

For any two states $x, y \in \mathcal{X}$,

$$\mathbb{P}\left(\{\text{state } y \text{ is visited exactly } k \text{ times}\} \mid \{X_1 = x\}\right) = \begin{cases} 1 - f_{xy}, & k = 0, \\ f_{xy} f_{yy}^{k-1} (1 - f_{yy}), & k \in \mathbb{N}. \end{cases}$$

Corollary 3

For any $x, y \in \mathcal{X}$,

$$\mathbb{E}\left[\underbrace{\#\text{visits to state } y}_{N_y} \mid \{X_1 = x\}\right] = \begin{cases} \frac{f_{xy}}{1-f_{yy}}, & f_{yy} < 1, \\ +\infty, & f_{yy} = 1, f_{xy} > 0. \end{cases}$$

Transience and Recurrence in a Nutshell

Theorem (Complete Characterisation of Transience and Recurrence)

Let $\mathbf{X} \stackrel{\text{MC}}{\sim} (\mu, P)$ be a homogeneous DTMC on a finite state space $\mathcal{X}$ with TPM P .

1. A state $y \in \mathcal{X}$ is transient if and only if: $\mathbb{E}[N_y \mid \{X_1 = y\}] < +\infty$.
2. A state $y \in \mathcal{X}$ is recurrent if and only if: $\mathbb{E}[N_y \mid \{X_1 = y\}] = +\infty$.

Remarks:

- Note that $N_y = \sum_{n \geq 2} \mathbf{1}_{\{X_n = y\}}$, and hence

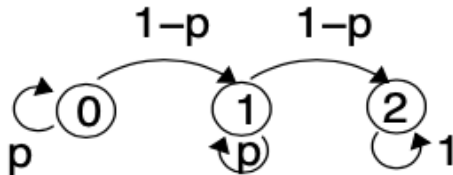
$$\begin{aligned} \mathbb{E}[N_y \mid \{X_1 = y\}] &= \mathbb{E}\left[\sum_{n \geq 2} \mathbf{1}_{\{X_n = y\}} \mid \{X_1 = y\}\right] \stackrel{\text{MCT}}{=} \sum_{n \geq 2} \mathbb{P}(\{X_n = y\} \mid \{X_1 = y\}) \\ &= \sum_{n \geq 2} P^{n-1}(y, y) = \sum_{n \in \mathbb{N}} P^n(y, y). \end{aligned}$$

Transience and Recurrence in a Nutshell

- A transient state y is visited only finitely many times a.s., no matter the initial state
- If y is recurrent and the Markov chain starts in y , then y will be visited infinitely many times a.s.
- If y is recurrent and the Markov chain starts in some initial state $x \neq y$ for which $f_{xy} = 1$, then y will be visited infinitely many times a.s.
- If $|\mathcal{X}| < +\infty$, then all states in $\mathcal{X}$ cannot be transient

Example

- Consider a time-homogeneous DTMC with following transition graph.



Compute f_{00} , f_{11} , and f_{22} .

When $p \in (0, 1)$, classify the states into transient, positive recurrent, and null recurrent.

Communicating Classes

Definition (Reachability)

A state $y \in \mathcal{X}$ is said to be **reachable** from state $x \in \mathcal{X}$ if there exists $n \in \mathbb{N} \cup \{0\}$ such that

$$P^n(x, y) > 0.$$

Notation: $x \longrightarrow y$.

Definition (Communication)

Two states x and y are said to **communicate** with each other if $x \longrightarrow y$ and $y \longrightarrow x$.

Notation: $x \longleftrightarrow y$.

Communication is an Equivalence Relation

Lemma (Communication is an Equivalence Relation)

The relation $\longleftrightarrow$ on $\mathcal{X} \times \mathcal{X}$ is an **equivalence relation**. Formally:

1. **(Reflexive)**: $x \longleftrightarrow x$ for all $x \in \mathcal{X}$.

2. **(Symmetric)**: For all $x, y \in \mathcal{X}$,

$$x \longleftrightarrow y \iff y \longleftrightarrow x.$$

3. **(Transitive)**: For all $x, y, z \in \mathcal{X}$,

$$x \longleftrightarrow y, \quad y \longleftrightarrow z \implies x \longleftrightarrow z.$$

Proof of Lemma 6

- Reflexivity:

$$P^0 = I \quad \Longleftrightarrow \quad P^0(x, x) = 1 > 0.$$

- Symmetry:

$$x \longleftrightarrow y \quad \Longleftrightarrow \quad x \longrightarrow y, \quad y \longrightarrow x \quad \Longleftrightarrow \quad y \longleftrightarrow x.$$

- Transitivity:

$$x \longleftrightarrow y, \quad y \longleftrightarrow z \quad \Longrightarrow \quad \exists m, n \in \mathbb{N} \cup \{0\} \text{ such that } P^m(x, y) > 0, \quad P^n(y, z) > 0$$

We then have

$$\begin{aligned} P^{m+n}(x, z) &= \sum_{w \in \mathcal{X}} P^m(x, w) \cdot P^n(w, z) \\ &\geq P^m(x, y) \cdot P^n(y, z) \\ &> 0, \end{aligned}$$

thus implying that $x \longleftrightarrow z$

Communicating Class

- The equivalence relation $\longleftrightarrow$ partitions $\mathcal{X}$ into equivalence classes
- These equivalence classes are called **communicating classes**
- A communicating class $\mathcal{C} \subseteq \mathcal{X}$ is said to be **open** if there exists an edge leaving $\mathcal{C}$ in the transition graph
Formally, there exists $x \in \mathcal{C}$ and $y \in \mathcal{C}^c$ such that

$$P(x, y) > 0.$$

- A communicating class that is not open is said to be **closed**
Formally, $\mathcal{C}$ is a closed communicating class if and only if

$$P(x, y) = 0 \quad \forall x \in \mathcal{C}, y \in \mathcal{C}^c.$$

Irreducibility and Periodicity

Irreducibility

Definition (Irreducibility)

A homogeneous DTMC $\mathbf{X}$ with TPM P is said to be **irreducible** if its entire state space constitutes a single communicating class.

Formally, for every choice of $x, y \in \mathcal{X}$, there exists $n \in \mathbb{N} \cup \{0\}$ (possibly depending on x, y) such that

$$P^n(x, y) > 0.$$

Remarks:

- Irreducibility is a property of the TPM, not of the process $\mathbf{X}$
- Some textbooks (particularly on RL) refer to irreducibility as **unichain** property
- If P is irreducible, there is a sufficiently large integer $N \in \mathbb{N}$ such that every entry of P^N is strictly positive

Period of a State

- For any $x \in \mathcal{X}$, let

$$\mathcal{R}(x) := \left\{ n \in \mathbb{N} : P^n(x, x) > 0 \right\}$$

denote the set of **return times** to x

Definition (Period)

The **period** of a state $x \in \mathcal{X}$, denoted $d(x)$, is defined as the GCD of the set of return times, i.e.,

$$d(x) := \gcd\{n \in \mathbb{N} : P^n_{x,x} > 0\}.$$

A state x is called **aperiodic** if $d(x) = 1$. A state x is said to be periodic if $d(x) > 1$.

Period is a Class Property

Lemma (Period is a Class Property)

If $x \longleftrightarrow y$, then $d(x) = d(y)$. Thus, all states within a communicating class possess the same period.

Proof:

- Observe that

$$x \longleftrightarrow y \implies \exists m, n \in \mathbb{N} \cup \{0\} \text{ such that: } P_{x,y}^m > 0, P_{y,x}^n > 0.$$

- Suppose $r \in \mathcal{R}(x)$, i.e., $P^r(x, x) > 0$. Then,

$$P^{n+m}(y, y) \geq P^n(y, x) \cdot P^m(x, y) > 0, \quad P^{n+r+m}(y, y) \geq P^n(y, x) \cdot P^r(x, x) \cdot P^m(x, y) > 0,$$

thereby implying that

$$n + m \in \mathcal{R}(y), \quad n + m + r \in \mathcal{R}(y).$$

- It then follows that

$$d(y) \mid n + m, \quad d(y) \mid n + m + r \implies d(y) \mid r.$$

Period is a Class Property

Lemma (Period is a Class Property)

If $x \longleftrightarrow y$, then $d(x) = d(y)$. Thus, all states within a communicating class possess the same period.

Proof:

- We have thus proved that if $r \in \mathcal{R}(x)$, then $d(y) \mid r$ (i.e., $d(y)$ divides every element of $\mathcal{R}(x)$)
That is, $d(y)$ is a common divisor of $\mathcal{R}(x)$
- This implies $d(y) \leq d(x)$, as $d(x)$ is the greatest common divisor of $\mathcal{R}(x)$
- Switching the roles of x and y , we can establish that $d(x) \leq d(y)$
- This proves that $d(x) = d(y)$

Transience and Recurrence are Class Properties

Theorem (Transience and Recurrence are Class Properties)

Transience and recurrence are class properties.

That is, the states within a communicating class are either all transient or all recurrent.

Proof:

- Suppose x is recurrent and $x \longleftrightarrow y$. Then, there exist $m, n \in \mathbb{N} \cup \{0\}$ such that

$$P^m(x, y) > 0, \quad P^n(y, x) > 0.$$

- We know that

$$x \text{ recurrent} \iff \mathbb{E} \left[N_x \mid \{X_1 = x\} \right] = \sum_{s \in \mathbb{N}} P^s(x, x) = +\infty.$$

Transience and Recurrence are Class Properties

Theorem (Transience and Recurrence are Class Properties)

Transience and recurrence are class properties.

That is, the states within a communicating class are either all transient or all recurrent.

Proof:

- We then have

$$\begin{aligned}
 \mathbb{E}[N_y \mid \{X_1 = y\}] &= \sum_{k \in \mathbb{N}} P^k(y, y) \\
 &\geq P^{m+n+1}(y, y) + P^{m+n+2}(y, y) + P^{m+n+3}(y, y) + \dots \\
 &= \sum_{s \in \mathbb{N}} P^{m+n+s}(y, y) \\
 &\geq \sum_{s \in \mathbb{N}} P^n(y, x) P^s(x, x) P^m(x, y) = P^n(y, x) P^m(x, y) \left(\sum_{s \in \mathbb{N}} P^s(x, x) \right) \\
 &= +\infty.
 \end{aligned}$$

Transience and Recurrence are Class Properties

Theorem (Transience and Recurrence are Class Properties)

Transience and recurrence are class properties.

That is, the states within a communicating class are either all transient or all recurrent.

Proof:

- Suppose x is transient, and $x \longleftrightarrow y$. Then, there exist $m, n \in \mathbb{N} \cup \{0\}$ such that

$$P^m(x, y) > 0, \quad P^n(y, x) > 0.$$

- We know that

$$x \text{ transient} \iff \mathbb{E}[N_x \mid \{X_1 = x\}] = \sum_{s \in \mathbb{N}} P^s(x, x) < +\infty.$$

Transience and Recurrence are Class Properties

Theorem (Transience and Recurrence are Class Properties)

Transience and recurrence are class properties.

That is, the states within a communicating class are either all transient or all recurrent.

Proof:

- We then have

$$\begin{aligned}
 \mathbb{E}[N_y \mid \{X_1 = y\}] &= \sum_{s \in \mathbb{N}} P^s(y, y) \\
 &= \frac{1}{P^m(x, y) P^n(y, x)} \sum_{s \in \mathbb{N}} P^m(x, y) P^s(y, y) P^n(y, x) \\
 &\leq \frac{1}{P^m(x, y) P^n(y, x)} \sum_{s \in \mathbb{N}} P^{m+s+n}(x, x) \\
 &\leq \frac{1}{P^m(x, y) P^n(y, x)} \sum_{k \in \mathbb{N}} P^k(x, x) < +\infty.
 \end{aligned}$$

Positive/Null Recurrence is a Class Property

Theorem (Positive/Null Recurrence is a Class Property)

Positive recurrence and null recurrence are class properties.

That is, the states within a communicating class are either all positive recurrent or all null recurrent.

Proof:

- For the proof, we will use the below result (a proof of which will be provided later)
- **Important result:**

Lemma (An Important Result)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n P^k(x, x) = \begin{cases} 0, & f_{xx} < 1, \\ \frac{1}{\mu_{xx} - 1}, & f_{xx} = 1. \end{cases}$$

- Using the above lemma, we first prove Theorem 13

Positive/Null Recurrence is a Class Property

Theorem (Positive/Null Recurrence is a Class Property)

Positive recurrence and null recurrence are class properties.

That is, the states within a communicating class are either all positive recurrent or all null recurrent.

Proof:

- It suffices to prove the lemma for null recurrence
- Suppose x is null recurrent and $x \longleftrightarrow y$. Then, there exist $r, s \in \mathbb{N} \cup \{0\}$ such that

$$P^r(x, y) > 0, \quad P^s(y, x) > 0.$$

- For any $n > r + s$, we have

$$\sum_{k=1}^n P^k(x, x) \geq \sum_{k=r+s+1}^n P^k(x, x) = \sum_{\ell=1}^{n-r-s} P^{r+\ell+s}(x, x) \geq \sum_{\ell=1}^{n-r-s} P^r(x, y) P^\ell(y, y) P^s(y, x)$$

Positive/Null Recurrence is a Class Property

Theorem (Positive/Null Recurrence is a Class Property)

Positive recurrence and null recurrence are class properties.

That is, the states within a communicating class are either all positive recurrent or all null recurrent.

Proof:

- For any $n > r + s$, we then have

$$\frac{1}{n} \sum_{k=1}^n P^k(x, x) \geq \frac{n - r - s}{n} \cdot \left(\frac{1}{n - r - s} \sum_{\ell=1}^{n-r-s} P^\ell(y, y) \right) \cdot P^r(x, y) P^s(y, x).$$

- Taking limits as $n \rightarrow \infty$ and using Lemma 14, we get

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{\ell \in \mathbb{N}} P^\ell(y, y) = 0.$$

Again, from Lemma 14, we get that y should be either transient or null recurrent

- But y cannot be transient, as from Theorem 9,

$$y \text{ transient} \implies x \text{ transient}$$